

Session-Based Counseling Chat System

<https://github.com/lyk4911-ship-it/Chat-agent>

Group# 10

Yukuo Li (yl11937)

Grace Smith(gs8717)

Ajit Sivakumar (as19268)

Date of Submission 5/7/26

Table of Work

(Please write x in the boxes to mention what each student achieved in this project)

	Yukuo Li	Grace Smith	Ajit Sivakumar
Project Description	x		
Uses Cases Diagram(s) and discription	x		
Sequence Diagrams	x		
Class diagram(s)	x		
Implementation	x		
Conclusion	x		

Written Project Description

General Description, Goals and Benefits

This project implements a session-based counseling chat system using object-oriented design principles. The system supports user registration/login, multi-session conversation management, real-time chat interaction, and basic administrative monitoring. From a user perspective, each account can create and manage multiple conversation sessions, continue previous discussions, and retrieve historical dialogue records. From an administrative perspective, the system provides session summaries, high-risk session filtering, and report-related functions for supervision use.

The primary goal of the project is to build a maintainable and extensible counseling dialogue platform that demonstrates clear OOP architecture in analysis, design, and implementation. To achieve this goal, the system is organized into layered modules: domain models, application services, infrastructure components, and HTTP interfaces. Core responsibilities are separated into dedicated classes (e.g., authentication, chat orchestration, session management, monitoring), improving readability, testability, and scalability.

The project provides several practical benefits. First, it improves conversation continuity through persistent session and history management. Second, it enforces clearer ownership and access control by validating user-session relationships before sensitive operations. Third, it enhances maintainability through abstraction-based design (interfaces and concrete implementations), allowing storage or API backends to be replaced more easily later on. Finally, it offers a structured foundation for future expansion, such as richer analytics, stronger role-based authorization, and database-backed persistence.

Special Requirements (Performance, Interfaces, Constraints, Reliability)

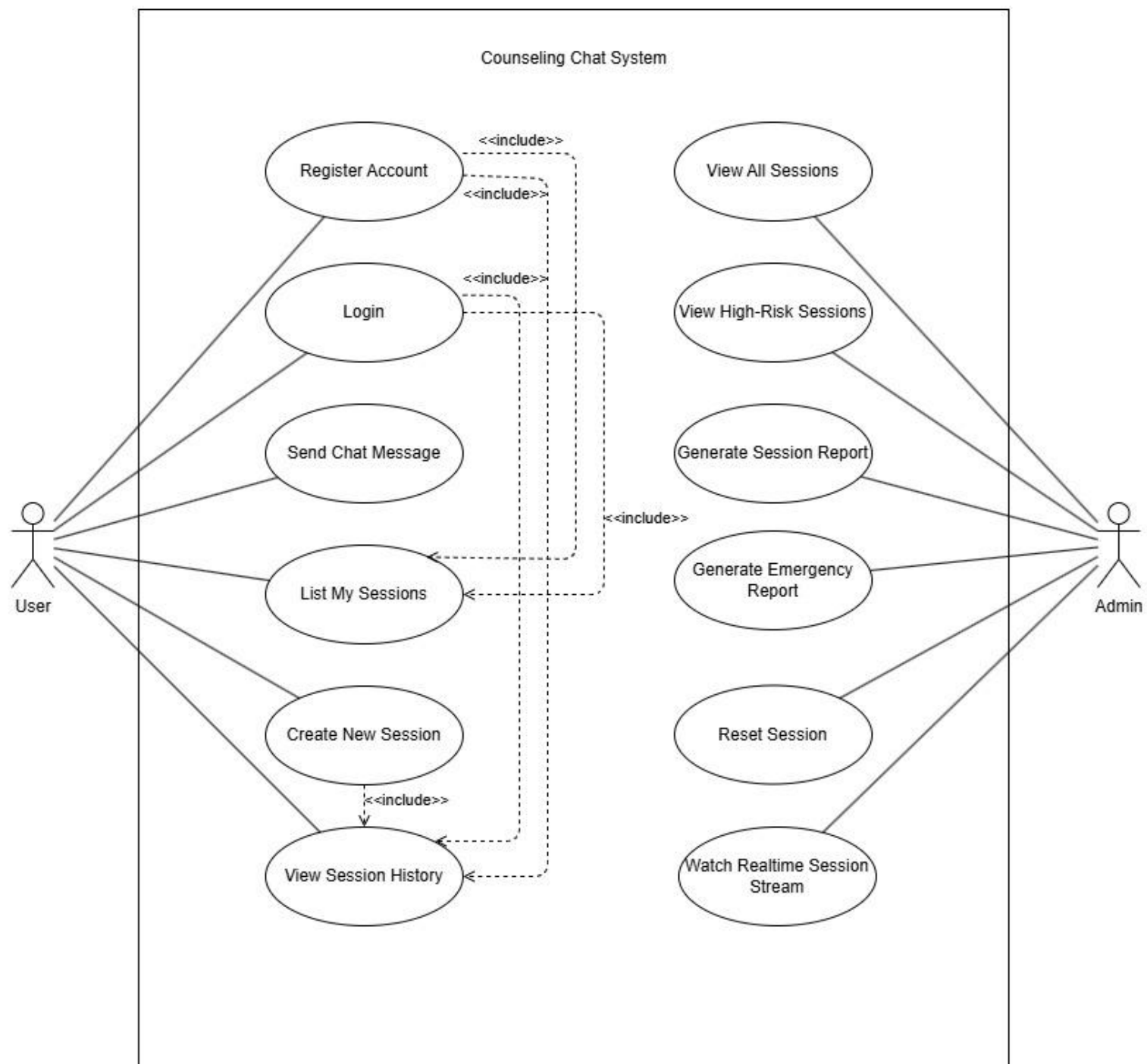
Performance: The system is designed for responsive interaction in small-to-medium course-project workloads. Session access is optimized through runtime caching/registry, while persistence operations use lightweight JSON/JSONL storage. Typical operations (login, listing sessions, retrieving history, sending messages) are expected to respond promptly under normal local deployment conditions.

Interfaces: The system exposes a REST-style HTTP interface (FastAPI) for authentication, chat, session management, and monitoring endpoints. A web entry route serves the chat UI. Request and response payloads are validated using schema models to keep interface behavior consistent and predictable.

Constraints: This implementation is intended as an educational OOP project and currently uses local file-based persistence (data/, logs/) instead of a production database. The quality and latency of LLM responses depend on external API availability and API key configuration. Some monitoring/report endpoints may require a tester token depending on runtime settings.

Reliability: The system includes structured error handling and maps service-level exceptions to appropriate HTTP status codes (e.g., 400/401/403/404). Input validation is enforced at the service and schema layers, and ownership checks are applied to prevent unauthorized session access. Automated tests (unit and HTTP smoke tests) are provided to verify core behavior and reduce regression risk.

Uses Cases Diagram



Use cases description

UC Reference Name/Number	UC-01 Register Account
Overview	A new user submits a username and password to create an account. The system validates the input, creates the user record, and returns initial session and history data for immediate use.
Related use cases	UC-04 List My Sessions, UC-06 View Session History
Actors	User

UC Reference Name/Number	UC-02 Login
Overview	An existing user logs in with a username and password. The system verifies the credentials and returns the user profile, active session, session list, and current session history.
Related use cases	UC-04 List My Sessions, UC-06 View Session History
Actors	User

UC Reference Name/Number	UC-03 Send Chat Message
Overview	A user sends a message to a selected chat session. The system validates the session, checks ownership when a user ID is provided, generates an assistant response, updates the session state, and records the interaction.
Related use cases	UC-02 Login, UC-05 Create New Session, UC-06 View Session History
Actors	User

UC Reference Name/Number	UC-04 List My Sessions
Overview	A user requests the list of chat sessions under their account. The system retrieves and returns session metadata such as session ID, title, creation time, and last update time.
Related use cases	UC-02 Login, UC-05 Create New Session, UC-06 View Session History
Actors	User

UC Reference Name/Number	UC-05 Create New Session
Overview	A user creates a new chat session with an optional title. The system validates the user, creates a new session ID, and returns the initial state and history bundle for that session.
Related use cases	UC-04 List My Sessions, UC-06 View Session History, UC-03 Send Chat Message
Actors	User

UC Reference Name/Number	UC-06 View Session History
Overview	A user retrieves the message history and current state of a specific chat session. The system validates the user, checks session ownership, and returns ordered conversation records with the latest state information.
Related use cases	UC-03 Send Chat Message, UC-04 List My Sessions, UC-05 Create New Session
Actors	User

UC Reference Name/Number	UC-07 View All Sessions
Overview	An admin requests an overview of all active sessions in the system. The system returns session summaries, including session identifiers, turn counts, state information, and risk-related indicators.
Related use cases	UC-08 View High-Risk Sessions, UC-09 Generate Session Report
Actors	Admin

UC Reference Name/Number	UC-08 View High-Risk Sessions
Overview	An admin reviews sessions that are marked with elevated risk signals. The system filters active sessions and returns high-risk session summaries for monitoring and follow-up.
Related use cases	UC-07 View All Sessions, UC-10 Generate Emergency Report
Actors	Admin

UC Reference Name/Number	UC-09 Generate Session Report
Overview	An admin generates a report for a selected session. The system collects session history, state information, and related interaction data, then returns a structured session report.
Related use cases	UC-07 View All Sessions, UC-08 View High-Risk Sessions, UC-06 View Session History
Actors	Admin

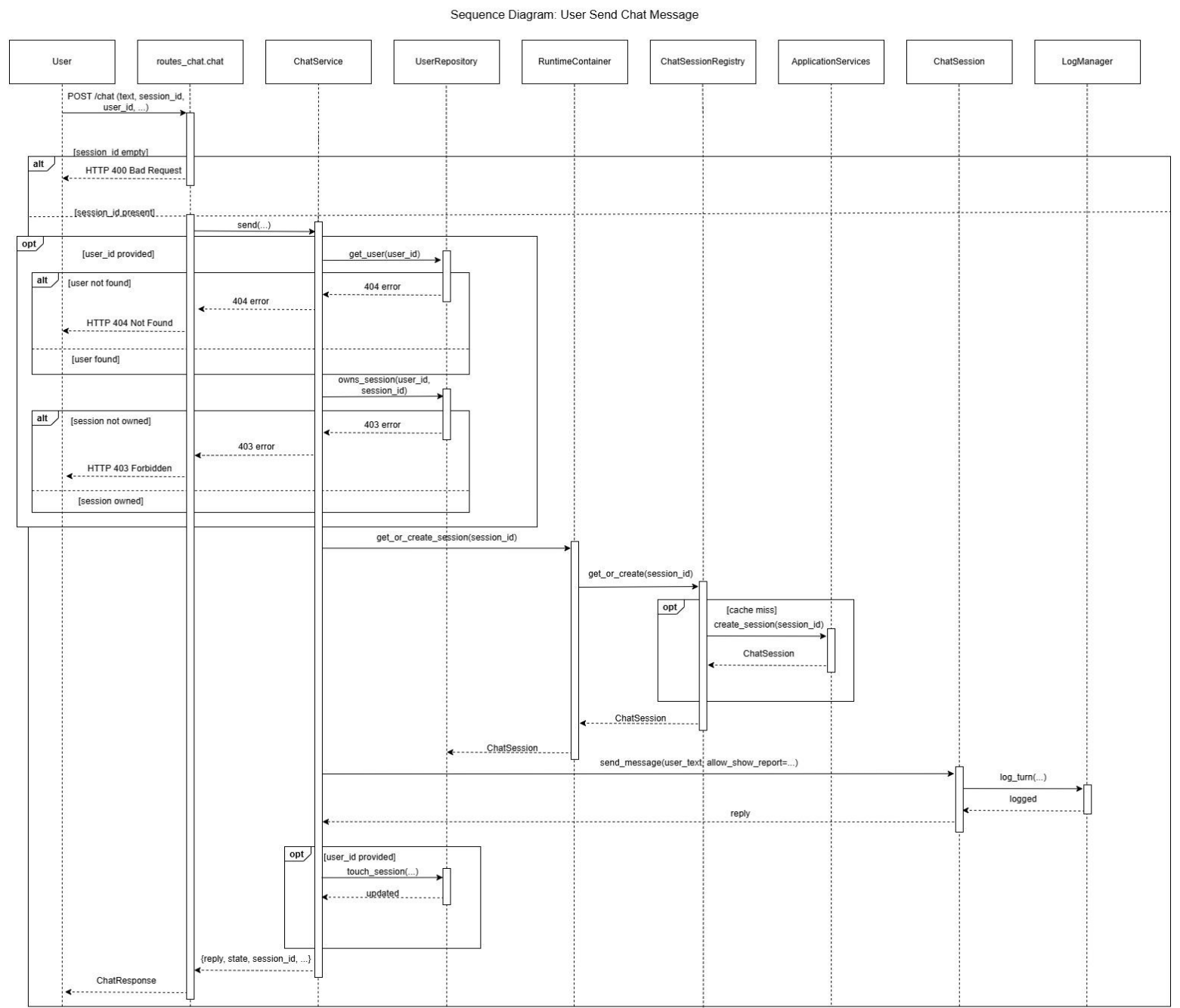
UC Reference Name/Number	UC-10 Generate Emergency Report
Overview	An admin generates an emergency report for a selected session when urgent review is needed. The system gathers risk-related session information and returns an emergency-focused summary.
Related use cases	UC-08 View High-Risk Sessions, UC-09 Generate Session Report
Actors	Admin

UC Reference Name/Number	UC-11 Reset Session
Overview	An admin resets a selected session. The system clears or reinitializes the runtime session state so that the session can start again from a clean state.
Related use cases	UC-07 View All Sessions, UC-09 Generate Session Report
Actors	Admin

UC Reference Name/Number	UC-12 Watch Realtime Session Stream
Overview	An admin watches real-time session updates through the monitoring stream. The system continuously provides live session information so the admin can observe ongoing activity and risk status.
Related use cases	UC-07 View All Sessions, UC-08 View High-Risk Sessions
Actors	Admin

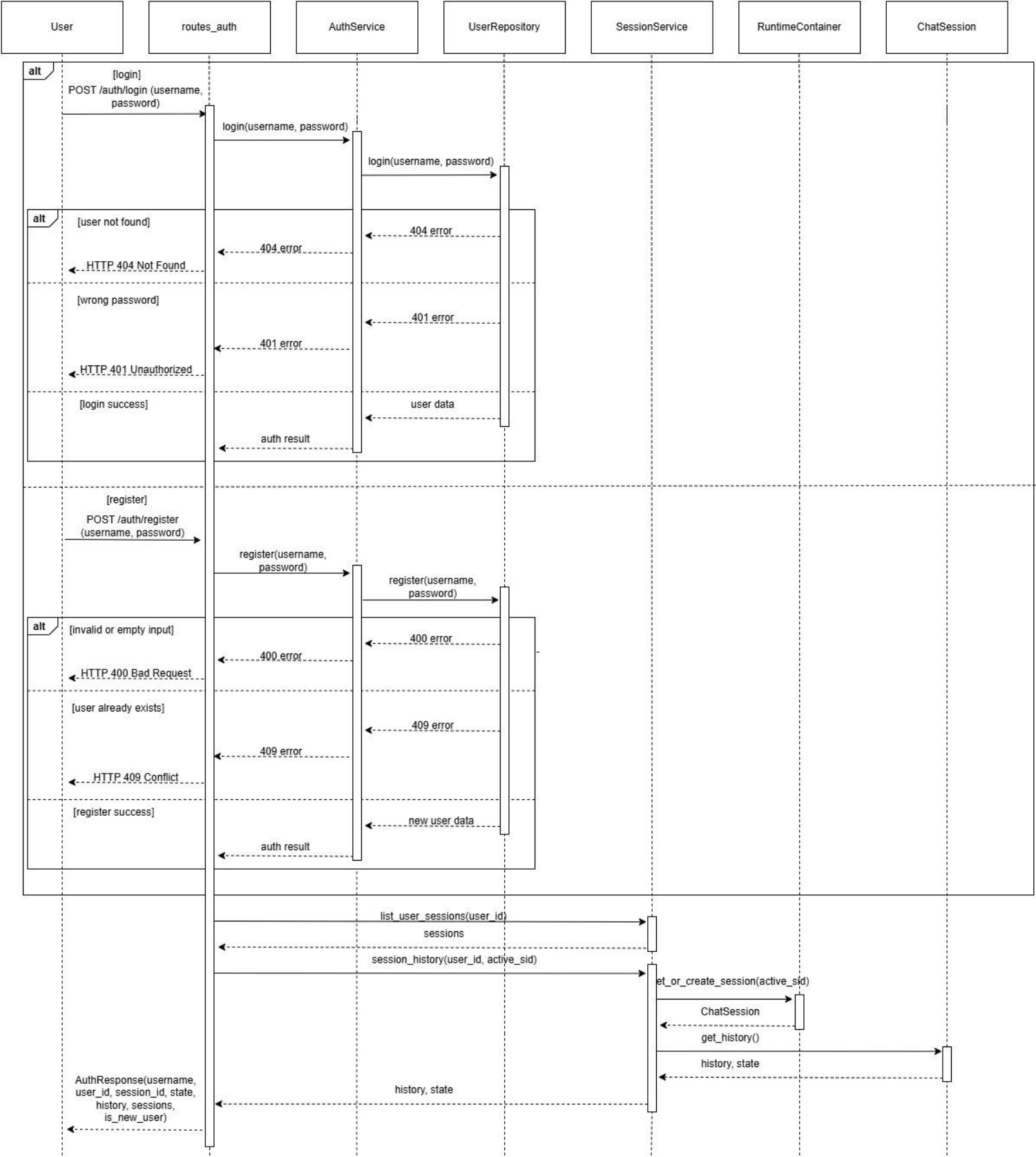
System Design

Sequence Diagram_ User Send Chat Message

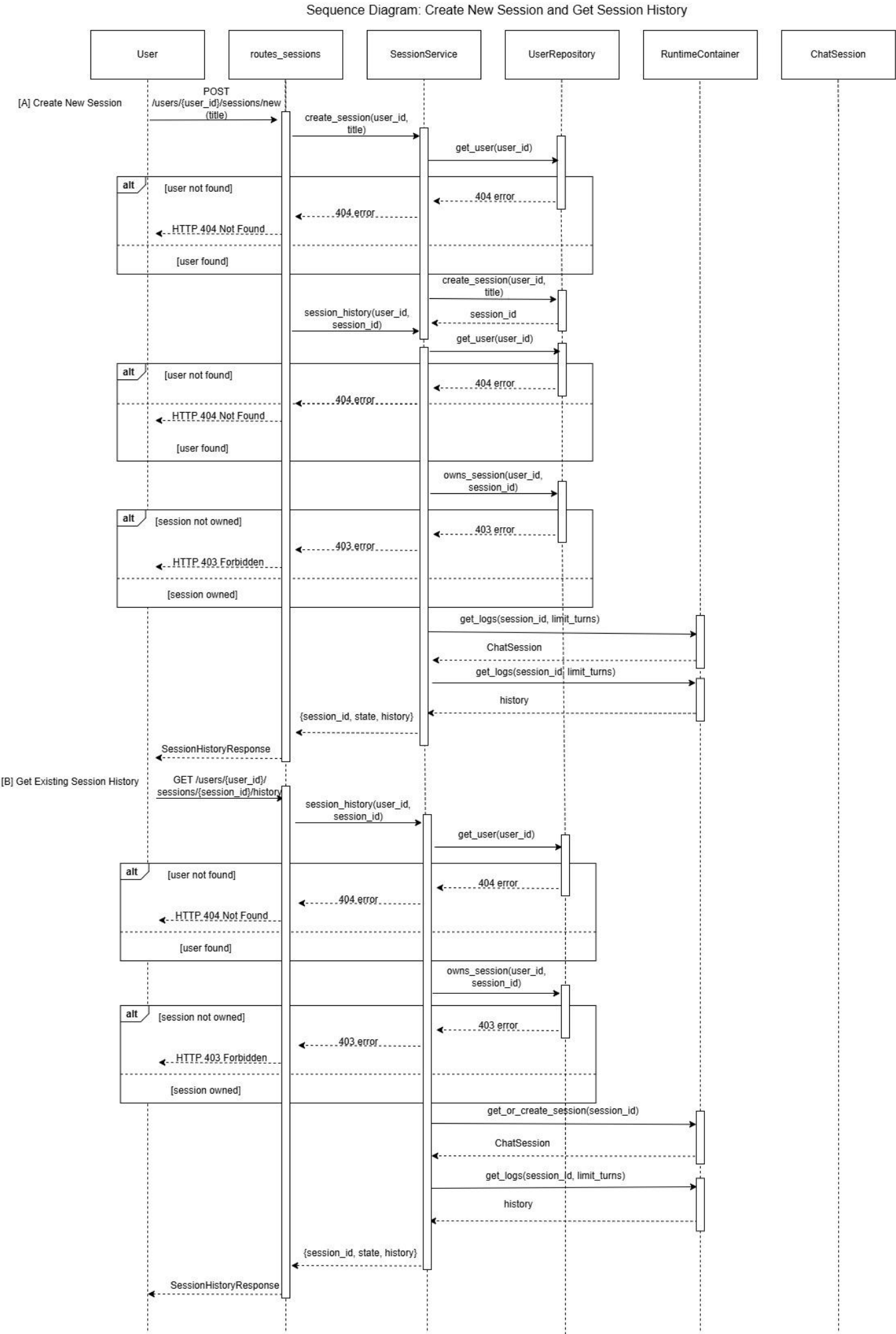


Sequence Diagram_ User Login _ Register

Sequence Diagram: User Login / Register

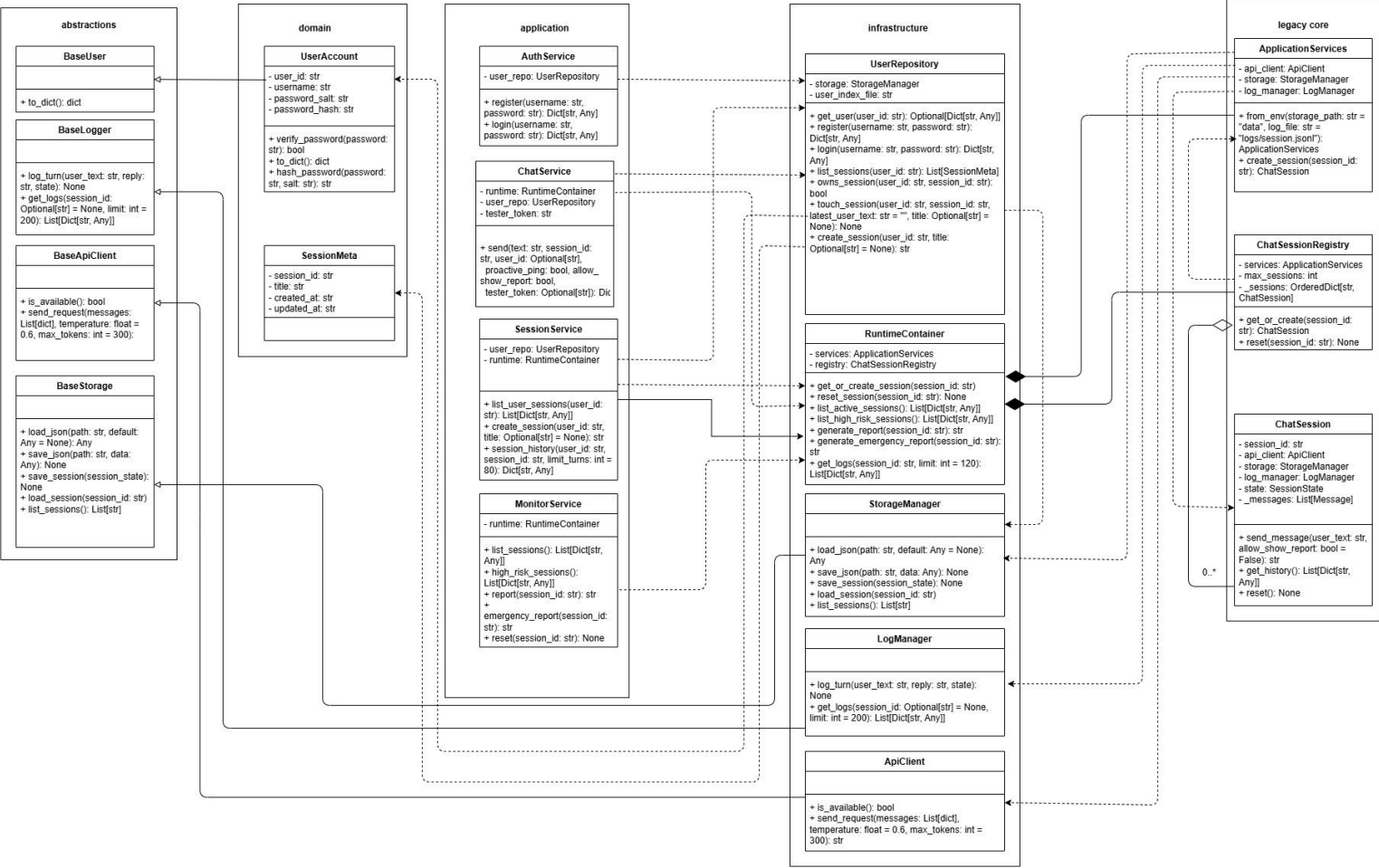


Sequence Diagram_ Create New Session and Get Session History



Class diagram

Class Diagram: Counseling Chat System



Conclusion (Work Summary, Difficulties, Recommendations)

This project delivered a complete session-based counseling chat system built on an object-oriented architecture. We implemented the core functional modules, including user registration/login, chat message processing, session creation and history retrieval, and admin-side session monitoring. The final design separates responsibilities across the domain, application, infrastructure, and interface layers, which improves maintainability and keeps the UML-to-code mapping clear and consistent.

During implementation, the main difficulty was maintaining design consistency while integrating legacy core components with the new layered structure. We addressed this by introducing clear service boundaries and runtime composition, so that the HTTP layer depends on application services while the existing chat engine components remain reusable. Another challenge was ensuring that session ownership validation and error mapping behaved consistently across multiple endpoints; this was handled through centralized service logic and structured exception handling.

For future improvement, we recommend migrating persistence from local JSON files to a database (e.g., SQLite/PostgreSQL) for stronger consistency and scalability, strengthening authentication and authorization (e.g., token-based auth), and extending monitoring and reporting capabilities with richer risk analytics. We also suggest adding more integration tests to cover edge cases and production-like scenarios.

Optional Appendix (Related Reports, Questionnaires, and Documents)

The appendix may include:

- Project proposal document (Yukuo_Li_Final_Project_Proposal.pdf)
- UML source files (draw.io / PlantUML)
- API endpoint summary (auth / chat / sessions / monitor)
- Test evidence (command and result screenshot, e.g., 97 passed)
- Environment and setup notes used for grading
- Additional design notes or architecture draft artifacts